

1 ILP Formulation

Let T be the set of tasks, J the set of jobs, C the set of cores, and K the set of clusters. Each job $j \in J$ belongs to a parent task $\tau(j) \in T$. Let $D_{\text{job}} \subseteq J \times J$ denote job-level precedence dependencies and $D_{\text{task}} \subseteq T \times T$ denote task-level communication dependencies. Each task $t \in T$ may only be assigned to an eligible core in $E_t \subseteq C$. For cluster $k \in K$, let $C_k \subseteq C$ denote its cores.

Model

$$\min \quad \lambda_{\text{core}} \sum_{c \in C} o_c^{\text{core}} + \lambda_{\text{cluster}} \sum_{k \in K} o_k^{\text{cluster}} + \sum_{(t_1, t_2) \in D_{\text{task}}} \sum_{c \in E_{t_1}} \sum_{d \in E_{t_2}} q_{cd} z_{t_1 t_2 cd} \quad (1)$$

Subject to:

Task assignment

$$\sum_{c \in E_t} y_{tc} = 1 \quad \forall t \in T \quad (2)$$

$$y_{tc} = 0 \quad \forall t \in T, c \in C \setminus E_t \quad (3)$$

Partitioned job assignment

$$x_{jc} = y_{\tau(j)c} \quad \forall j \in J, c \in C \quad (4)$$

Timing

$$s_j \geq r_j \quad \forall j \in J \quad (5)$$

$$s_j \leq r_j + \text{jitter} \quad \text{for applicable root jobs} \quad (6)$$

$$f_j = s_j + \sum_{c \in C} p_j \alpha_c x_{jc} \quad \forall j \in J \quad (7)$$

$$f_j \leq d_j \quad \forall j \in J \text{ with } d_j \neq \emptyset \quad (8)$$

Precedence

$$s_{j'} \geq f_j \quad \forall (j, j') \in D_{\text{job}} \quad (9)$$

Core memory

$$\sum_{t \in T} m_t y_{tc} \leq B_c^{\text{core}} + o_c^{\text{core}} \quad \forall c \in C \quad (10)$$

Cluster memory

$$\sum_{t \in T} \sum_{c \in C_k} m_t y_{tc} \leq B_k^{\text{cluster}} + o_k^{\text{cluster}} \quad \forall k \in K \quad (11)$$

Non-overlap on cores

$$s_{j'} \geq f_j - M(3 - u_{jj'} - x_{jc} - x_{j'c}) \quad \forall j, j' \in J, j < j', c \in C \quad (12)$$

$$s_j \geq f_{j'} - M(2 + u_{jj'} - x_{jc} - x_{j'c}) \quad \forall j, j' \in J, j < j', c \in C \quad (13)$$

Communication linearization

$$z_{t_1 t_2 cd} \geq y_{t_1 c} + y_{t_2 d} - 1 \quad \forall (t_1, t_2) \in D_{\text{task}}, c \in E_{t_1}, d \in E_{t_2} \quad (14)$$

Domains

$$y_{tc} \in \{0, 1\} \quad \forall t \in T, c \in C \quad (15)$$

$$x_{jc} \in \{0, 1\} \quad \forall j \in J, c \in C \quad (16)$$

$$u_{jj'} \in \{0, 1\} \quad \forall j, j' \in J, j < j' \quad (17)$$

$$z_{t_1 t_2 cd} \in \{0, 1\} \quad \forall (t_1, t_2) \in D_{\text{task}}, c \in E_{t_1}, d \in E_{t_2} \quad (18)$$

$$s_j, f_j \in [0, M] \quad \forall j \in J \quad (19)$$

$$o_c^{\text{core}} \geq 0 \quad \forall c \in C \quad (20)$$

$$o_k^{\text{cluster}} \geq 0 \quad \forall k \in K \quad (21)$$